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भारतीय भूवैज्ञानिक सर्वेक्षण GEOLOGICAL SURVEY OF INDIA

जम्मू एवं कश्मीर के ऊधमपुर जिला के खेरी में भूस्खलन पर विस्तृत जांच रिपोर्ट
मद कूट: ई.जी./सी/एनआर/जेके/2015/049
(कार्य सत्र: 2015-16)

**REPORT ON DETAILED LANDSLIDE INVESTIGATION AT KHERI, UDHAMPUR
DISTRICT, JAMMU & KASHMIR**
Item Code: EG/C/NR/JK/2015/049
(Field Season: 2015-16)

द्वारा
देवेन्द्र सिंह, वरिष्ठ भूवैज्ञानिक
एवं
सुरेश कुमार राणा, वरिष्ठ भूवैज्ञानिक

By
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द्वारा

देवेन्द्र सिंह एवं सुरेश कुमार राणा, वरिष्ठ भूवैज्ञानिक
भारतीय भूवैज्ञानिक सर्वेक्षण

सारांश

भारतीय भूवैज्ञानिक सर्वेक्षण ने प्रभावित क्षेत्र का विस्तृत भू-तकनीकी आँकलन किया और उपचारात्मक सुझाव दिये । राष्ट्रीय राजमार्गों के भू-स्खलन विभिन्न रूपों में घटित होते हैं और मोड़ पैमानों एवं आवृत्ति में परिवर्तित होते हैं । बदलाव भूगर्भिक संरचना द्वारा नियंत्रित होते हैं। इसलिये विभिन्न भागों के विस्तृत अध्ययन की आवश्यकता पड़ी जो कि तलछटी वातावरण व रुपान्तरित इलाकों की भूस्खलन में भिन्नता करने के लिये आवश्यक है। भूविज्ञान द्वारा भूस्खलन पर मुख्य कारकों को समझने का मुख्य उद्देश्य यह है कि भविष्य में भूस्खलन की घटना की भविष्यवाणी और भूस्खलन की अस्थिरता की पहचान की जा सके । कारणों/कारकों की पहचान जो कि निश्चित भूगर्भिक संरचना से पूर्व चेतावनी या कमजोर ढलानों की पहचान करने में एक बड़ा कदम हो सकता है । उद्देश्यों को प्राप्त करने के लिये स्थलाकृतिक नक्शों एवं क्षेत्र सर्वेक्षण की मदद से अध्ययन क्षेत्र की विस्तृत मैपिंग की गयी । खैरी, ऊधमपुर जिला, जम्मू एवं कश्मीर की विस्तृत भूस्खलन जाँच कार्यसत्र : 2015–16 में (मद कूट EG/C/NR/ JK/2025/049) भारतीय भूवैज्ञानिक सर्वेक्षण के तहत की गई ।

भूगर्भिक तौर पर क्षेत्र वलयदार और बैंगनी से ईट जैसे लाल भोल, सिल्टस्टोन और मिट्टी के पत्थरों का वैकल्पिक अनुक्रम जो कि भुरभुरा और कम ठोस चट्टान समूह, बफबैंगनी कठोर के साथ अंतरपरतीय, ठोस और मुरे वर्ग चट्टान का खण्डित बलुआ पत्थर को दर्शाता है । इस तरह की चट्टानें ढलान की ओर पूरी तरह से दिखती है या पायी जाती हैं। चट्टानों की दिशा $N60^{\circ} W - S 60^{\circ} E$ से $N70^{\circ} E - S 70^{\circ} E$ तक बदलती है। खैरी भूस्खलन मरी वर्ग की चट्टानों में घटित होती है । प्रभावित क्षेत्र का भूतकनीकी आँकलन किया गया और भूस्खलन प्रभावित क्षेत्रों उपचारात्मक सुझाव दिये गये जो कि इस बात पर

निर्भर करता है । कि जलनिकासी को ठीक किया जाये, सड़क के स्तर दो कटान से बच जाना चाहये, भूस्खलन के कई स्तरों पर बेंचों का निर्माण, वनस्पति का संरक्षण और नालों के समुचित बहाव जो की आवश्यक है, पर निर्भर करता है ।

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ABSTRACT

Geological Survey of India has conducted detailed geotechnical assessment of the affected area and suggested remedial measures. Landslides along National highways occur in various forms and vary in mode, scale and frequency. The variations appear to be significantly controlled by geologic setting, hence the need to study in detail the distinctive features that differentiate landslides in sedimentary environments from those on metamorphic localities. The aim is to understand the actual features impacted on the landslides by geology, from which future predictions of occurrence and identification of instability could be based. The recognition of features that are characteristic of certain geologic setting may prove to be a major step in early warning or identification of vulnerable slopes. To achieve the objectives detailed mapping of the study area was carried out using topographic maps, and field survey. The detailed landslide investigation at Kheri Udhampur Districts, Jammu & Kashmir was carried out in pursuance to the Geological Survey of India field season programme under Item code no. EG/C/NR/JK/2015/049 for Field Season 2015.

Geologically, the area consists of folded and alternate sequence of purple to brick red shale, siltstone, and clay stone representing fragile and poorly consolidated rock mass, inter-bedded with buff purple hard, compact and fractured sandstone of the Muree Group of rocks. These rocks are well exposed along the slope as scanty patches and occur prominently at the ridges. The trend of the rock varies from N60°W – S60° E to N70° W – S 70° E with 50° to 60° northerly dips. The Kheri slide has occurred within Muree Group of rocks. Detailed geotechnical assessment of the affected area was done and suggested remedial measures at landslide affected site as under, drainage needs to be channelized, Toe cutting at road level should be avoided, creation of benches along multiple locations of slide, Preserving vegetation and Proper training of nala is required.

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I. INTRODUCTION

The incessant rains (August second week 2015) caused natural calamities in the Udhampur District including many Landslides along National Highway-44 connecting Kashmir to the rest of country. During rainy season the Highway at Kheri gets blocked due to landslide at many times throughout the year disrupting traffic on this important National Highway.

As per instructions from Sh. Brij Kumar, Deputy Director General a team of Geologists comprising S/Shri Devender Singh, Suresh Kumar Rana, under the supervision of Sh.S K Tripathi, Director, carried out detailed geotechnical studies in the area. The authors have carried out detailed landslide investigation at Kheri in Udhampur Districts, Jammu & Kashmir in pursuance to the Geological Survey of India field season programme 2015-16 under Item code no. EG/C/NR/JK/2015/049.

Major drainage in the area is Tawi River which is a principal tributary of the Chenab River flowing through its left banks. The source of the Tawi River is closely located to the Kali Kundi glacier and adjoining area southwest of Bhadarwah in Doda district. Its catchment is bounded by latitude 32°35'-33°5'N and longitude 74°35'-75°45'E.

The catchment area of the river up to Indian border (Jammu) is 2168 km² and falls in the districts of Jammu, Udhampur and a small part of Doda. Elevation in the catchment varies between 400 and 4000 m. The flow of water in the Tawi River has been decreasing in recent years as the source glacier (Kali Kundi glacier) has been retreating.

The Udhampur district has a total geographical area of 4540 sq km. In 2007, Udhampur district has been divided into two districts Reasi and Udhampur. The Udhampur district has the distinction of having holy Mata Vaishnoo Devi Shrine. Patnitop and Kud are the summer hill resorts of the district where as the Mansar Lake enhances the beauty of the district.

1.1 LOCATION AND ACCESSIBILITY:

The area falls in survey of India Toposheet No.43P/1. Udhampur is well connected by road as it is located on National Highway 1A leading to Srinagar, which is the only national highway that connects Srinagar to the rest of India. From Udhampur the Kheri is around 5.0 km on NH-44 (Fig.01). Udhampur lies between latitude $32^{\circ} 39' 0''$ to $33^{\circ} 31' 20''$ N and longitude $74^{\circ} 35' 20''$ to $75^{\circ} 40' 30''$ E.



Fig.01: Location map of Udhampur.

1.2 HAZARDS, RISK AND VULNERABILITY OF J&K STATE:

The state of Jammu and Kashmir is very distinct from the rest of the country with respect to topography, climate, economy, social setting and strategic location. The state is a multi hazard prone region with natural disasters like earthquakes, floods, landslides, avalanches, high velocity winds, snow storms, besides manmade disasters including road accidents and fires etc. occurring in various parts of the state. Table 1 gives the statistical information related to the number of deaths due Landslide in the State. It also shows that in terms of loss of life due to landslide, the state contributes a significant share to the national average. Over the years from 2011 onwards the loss of life significantly increases in this part due to poor understanding of

landslides contributed by heavy downpour (September 2014) vis a vis settlement/relocation of affected people and unscientific cutting of roads, and building/laying of heavy civil engineering structures without proper geotechnical input.

Table 1 Number of Deaths due to Landslides in Jammu and Kashmir.

Year	2011	2010	2008	2007	2005	2004
Male	4	11	18	7	13	12
Female	3	1	1	0	3	1
Total	7	12	19	7	16	13
% share w.r to All India	2.2	2.1	6.2	2.5	1.4	8.6

Source: Compiled from the statistics released by Ministry of Home Affairs, Govt. of India

AREAS PRONE TO LANDSLIDES IN J&K STATE:

Areas along major highways particularly Udhampur to Ramban, Panthial, Banihal, Doda, Kishtwar, Gulmarg, Dawar, Gurez, Tangdhar, Rajouri etc. are landslide prone areas.

1.3 GEOLOGY OF STUDY AREA:

The dominant rock types in Kheri area are sandstone, red coloured clay, shale and siltstone that range from the Miocene to Oligocene in age and were deposited in environments that pass from continental in the proximal parts through paralic to shallow marine in the distal parts.

(a) FLORA AND FAUNA:

The region reveals varied vegetation that profoundly grows due to its rich soil types. Adequate rains coupled with a favourable salubrious climate also assist to retain the lushness of the forests and landscapes which is home to flourishing wildlife habitat. The flora of Jammu ranges from the thorn bushes of arid plains to moderate alpine of higher altitudes including Horse, chestnuts, maple, and silver fir that are broad leaf trees as well as Birch, berbers, rhododendrons and herbal plants which form a part of the high altitude region. In Poonch and Udhampur, hilly regions also feature a large varied population of fauna that include the bear, brown musk shrew, cheetah (*Acinonyx jubatus*), deer, leopards, muskrat, and wild sheep, including a variety of reptiles, bats, snakes, frogs, and lizards.

(b) PHYSIOGRAPHY AND DRAINAGE:

The districts have distinct topography. It is interwoven with several hill ranges and mountains. There are a number of intermountain valleys in the area. The altitude variation of the study area ranges from approx. 600 m to 3000 m. The area under study has rugged topography. The elevation, however, gradually rises from about 610 metres (2000 ft.) in the valleys in the southern portion to about 1829 meters (6000 ft.) above mean sea level in the northern portions of the area and to greater elevation in the main Pir Panjal Range further to the north. The valleys are generally narrow and deep, and swift currents with steep gradient are present in the biggar nalas. The major part of the area is occupied by Murees and Siwaliks. The area is drained by major River Tawi. The drainage network is mainly dendritic, dichotomous and trellis pattern. The investigated area is mountainous, precipitous and rugged with steep gorges and rocky uplands. The altitude of Udhampur ranges from 755 m to 2024 m (at Patnitop which is around 50 km from Udhampur town) above mean sea level.

The district is drained by a number of perennial rivers and ephemeral streams. They form part of Indus major basin and Chenab sub-basin. Chenab River and its tributaries viz., Tawi River, Basantar River and Ujh River (tributary of Ravi) are perennial rivers draining the district. These streams originate from northern mountainous regions and flow in south-westerly direction. These rivers are fed by number of first to fourth order streams in the study area. Number of local streams or nalas remain dry in summer but create havoc due to flash floods especially in rainy seasons. They carry huge load of boulders, pebbles, sand and silt during monsoon period.

(c) CLIMATE

Due to variation in altitude from 755 m to 3000 m meters above mean sea level, there is a wide variation in climatic conditions in different parts of the district, experiencing a typical temperate climate in high altitude with snowfall and cold winter; whereas tropical climate at low altitude. The summer season starts from April to June followed by rainy season from July to September. The October and November although generally dry, are most pleasant months of the year. The winter season begins from December and ends in March. The high altitude areas experience very cold winter and mild summer. The temperature in the snowfall zone varies between sub-zeros to about 35°C. Sub-tropical climate prevails in the low altitude area where there is cold winter but scorching summers. Normal annual rainfall at Udhampur is 1400 mm. Mostly rainfall (~70% of the total annual rainfall) is received from southwest monsoon. Rainfall is also received during winter season due to western disturbances.

1.4 ACKNOWLEDGEMENT:

The authors are grateful to Shri Brij Kumar, Dy. Director General SU: Jammu, J&K, for extending support and providing logistics as well as guidance in accomplishing this work. It is exclusively a unique opportunity to express my deepest sense of gratitude to my esteemed supervisor Shri S.K Tripathi, Director, and Supervisory officer, Engineering Geology Division for overall supervision, technical guidance during the Field work and preparation/scrutiny of the report. Thanks are due to Shri Vijay Kapoor Sr. Surveyor GSI, Chandigarh for carrying out survey/ mapping during the course of study. I express my appreciation to Mr. Imran Khan Geologist for valuable cooperation during preparation of this report and people of Kheri village for extending necessary cooperation during field work.

1.5 PREVIOUS WORK:

Medlicott (1876) while dealing with the geology of the Tertiary belt of the north western part of J&K State classified 'Murree series' into Lower and Upper group comprising shale, sandstone and pseudo- conglomerate, and shale and sandstone, respectively. Singh (1963- 64, 64- 65) carried out systematic geological mapping of Murrees in parts of Poonch and Udhampur district and suggested that the Murree rocks be named as Murree Mollasse deposits or Formation as that would convey the nature of lithology, environment and site of deposition and structural position of these formations. Sharma (1964-65) carried out systematic geological mapping in parts of Poonch and Udhampur districts. Bhamrah (1967- 68) mapped parts of Kathua and Udhampur districts. He described the Lower Murrees as a sequence of alternating bands of sandstone, shale, siltstone, and pseudo-conglomerate, exhibiting lateral variation. Sandstone is generally hard and fine grained and contained impressions of dicotyledonous Angiosperms of Lower Miocene age. Conformably overlying the Lower Murrees, the Upper Murrees consist of softer and dominantly silty stage i.e. sandstone bands with alteration of clay bands and pseudo- conglomerate. Bharat Bhushan carried out sedimentological studies of Murree and Siwalik sediments for demarcating the boundary in parts of Kathua and Udhampur districts during the F.S. 1968- 69, 1969-70 and 1970- 71. The studies carried out by him revealed that the Murrees differ from the Kamlials in average grain size diameter and grain size distribution of sand in the sandstone, and in the abrasion index and degree of elongation of zircons. The Palaeo-current data, however, indicated that both the Murrees and Kamlials are not differentiable. In the absence of any field parameter to distinguish the two formations from each other, he recommended that these be grouped together for the purpose of field mapping and

Murree- Siwalik boundary be shifted to Kamliyal - Chinji boundary. He, however, advocated a palaeontological approach for the solution of this problem. Oil and Natural Gas Commission has been actively engaged in carrying out the systematic geological mapping and sedimentological studies of the Tertiary rocks of this belt ever since its inception in late fifties. Recognising the various rock types as rock- stratigraphic Units, they also divided the Murrees into Lower and Upper Murree formations. The Lower Murree is mainly argillaceous and contains thick, deep purple to chocolate coloured shale and siltstone. Sandstone is subordinate. The shale is highly veined with calcite. The siltstone is purple to chocolate in colour, hard and micaceous. The sandstone is fine grained well bedded, flaggy, and hard. Intraformational conglomerate is common. The Lower Murrees contain pebble beds in its top part as observed in the northern part of this belt i.e. nearer to the Main Boundary Fault. Based on the composition of these pebble beds, which are dominantly made up of sandstone, siltstone and minor amount of quartzite and limestone, they have advocated a local uplift and erosion of the Lower Murree Formation. The Upper Murree is more arenaceous but the purple colour is still dominant. Compared to the Lower Murree, the Upper Murree are less disturbed and characterized by ridge forming sandstones. The clays are dominantly purple while greenish grey types are subordinate.

1.6 PRESENT WORK:

An area of 0.06 sq km was studied for detailed landslide investigation from 22nd January to 04th February 2016 on 1:1000 scale and prepared a detailed geological map (Plate, I) and section along and across slide (Plate.II,III). The present work deals mainly with site specific detailed geotechnical investigation of Kheri landslide to unravel the possible causes and recommendations/remedial treatments. The work was completed in twenty eight man days. This report deals with the description of the work based on field evidences and interpretation of data thereon.

II REGIONAL GEOLOGY:

The area of investigation exposes oldest rocks of Sirban Group represented by dolomite, cherty dolomite with occasional slate/quartzite bands. These rocks form inlier known as Vaishnodevi Inlier which is surrounded by Murree Group represented by claystone, sandstone and siltstone. Towards south of the area, northern boundary of Murree Group is marked by Reasi Thrust which has brought Sirban Group over the Murree Group. The northern boundary of Murree Group is marked by Murree Thrust which has brought a narrow, complex belt of volcano-sedimentary sequence of autochthonous nature over it. The northern boundary of the autochthonous belt of the Lesser Himalaya is marked by the Panjal Thrust which has brought the entire Tethyan Himalayan Phanerozoic succession of Kashmir with its crystalline basement over it. Panjal Volcanic Formation is exposed along Banihal pass/tunnel. Towards north of Banihal Tunnel the folded Triassic limestone along with minor Jurassic shale beds are exposed in the section upto Qazigund. Quaternary sediments and Karewa rocks are intermittently exposed from Qazigund to further north upto Srinagar. The lithostratigraphic succession in order of superposition as worked out by the officers of Geological Survey of India and published in GSI, Miscellaneous Publication No.30, Part-X is given in Table-2.

Table-2: Lithostratigraphic succession

Super Group/ Group/Formation	Geological age	Lithology
Quaternary	Recent	Slope debris/scree/terrace deposits and re-cemented limestone scree
Siwalik	Miocene- Pliocene	Sand rock, conglomerate, sandstone, clay and shale
Murree	Miocene-Oligocene	Sandstone, shale, claystone and siltstone with calcite veins
Subathu	Eocene	Carbonaceous shale, olive green and purple thinly bedded splintery shale, coal bands, shelly limestone and nummulitic limestone
Jangalgali	Cretaceous-Eocene	Bauxite, aluminous clay, greyish thinly bedded quartzite and quartzitic breccia
Zilant	Neoproterozoic	Agglomeratic slate, tuff, phyllite, quartzite and limestone
Ramban	Neoproterozoic	Phyllite, slate, quartzite with gypsum bands
Ramsu	Neoproterozoic	Diamictite, slate, phyllite, quartzite, schist and limestone
Salkhala	Proterozoic	Carbonaceous phyllite, schist, gypsum limestone and marble
Sirban	Proterozoic	Light grey and grey quartzite, slate, massive dolomite, non-cherty dolomite, stromatolitic limestone and flaggy limestone
<u>INTRUSIVES</u> Granite gneiss		

Source: Miscellaneous Publication No.30.

III GEOLOGY OF KHERI SLIDE:

The slope on which the slide is located is mostly occupied by talus and scree of purple and buff sandstones and purple and red shale and clays of Murree group of Lower Miocene to Upper Eocene age. Along the slide portion fairly big boulders of Murree sandstone are also seen. Outcrops of these rocks along the slope are few but are prominent on the ridges to the east and the west where they are seen to be in the form of buff and purple sandstones and purple to brick red shale and clay. The material involved in sliding includes shale, soil, and rocks of Lower Murree group having boulder size ranging from 0.5 to 5.0 m diameter.

IV GEOTECHNICAL ASSESSMENT:

Detailed geotechnical assessment of the landslide area was done and remedial measures were suggested. Geologically, the area consists of folded and alternate sequence of purple to brick red shale, siltstone, and clay stone representing fragile and poorly consolidated rock mass, inter-bedded with buff purple hard, compact and fractured sandstone of the Murree Group of rocks. These rocks are well exposed along the slope as scanty patches and occur prominently at the ridges. The trend of the rock varies from N60°W – S60° E to N70° W – S 70° E with 50° to 60° northerly dips. The Kheri slide has occurred within Murree Group of rocks.

The team carried out detailed studies of Kheri landslide and its flank area along NH-44 in Udhampur district. The terrain is undulatory in nature with small mounds and depressions and at places cut by deep nalas. The rocks in the landslide zone can be divided into three zones based on the intensity of weathering: slight (Toe), moderate (Middle), and severe (Crown). The dimensions of slide are: height is 427.0m, width average 100.0 m at crown level at 1099.0m and bottom at 674.0m.

The sandstone exposed in the form of ridges/escarpment is jointed in nature. Numerous sets of joints are present in the more competent sandstone and siltstone beds of the Murree Formation. Most of the joints are dipping at 55° to 85° either towards NE or SW. The following are the prominent sets of joints recorded in the various rock types of Murree Group in the area: i) N 35°W-S35°E with 55° to 85° dip due southerly. ii) N70°W-S70°E with 65° to 80° dip due northerly. iii) N35°E-S35°W with 60° to 80° dip due southerly. iv) N55°E-S55°W with vertical dips v) N80°E-S80°W with 70° to 80° due northerly.

During the course of field work it was ascertained at field that following factors contributed towards failure of slopes in the studied area.

(1) Geomorphic setting and environment like, total topographic relief of area, steepness of hill slopes, less vegetation cover, slope orientation, climate and antecedent moisture condition in the area.

(2) External contributing factors Such as (a) excessive precipitation during the month of August second week 2015 (b) Anthropogenic activity at toe level (repeated clearance of slided material to make space for road).

(3) Changes in the slope height as a result of vertical erosion by Nalla, resulting into release of lateral stresses.

(4) Shock and vibration produced by large-scale explosion during cutting of roads for expansion of National Highway-44 and tunnelling work at Nashri might have gradually affected the equilibrium of slope over an extended period of time, on account of temporary changes of stress that are caused by oscillations of different frequencies. In loose sand rocks, shocks may disturb inter-granular bonds and thus decrease cohesion. In water saturated fine sand and sensitive clays, the displacement or rotation of grains can result in a sudden liquefaction of soil.

(5) Rain water penetrates into joints at Crown level and produces hydrostatic pressure. In soils the pore water pressure increases and consequently the shear resistance decreases. Clayey soils are desiccated and shrunken so that water readily percolates deep into the fissures. Swelling pressure and pore water pressure may also destabilize such material on the slopes (such as at Kheri).

(6) The clay minerals have significantly reduced the rock strength and facilitated the extensive failure of Kheri Slide. Clay-rich debris is also responsible for the formation of debris flows in the monsoon season.

Further, it is noteworthy to mention here that the whole of J&K is comes under Zone-V of Earthquake Zonation Map of India and one major tectonic plane, viz. MBT (Reasi Thrust) is passing through the northwest side of the area, not very far away from the site. It was also observed that another thrust is present about 3 km north of the slide.

TYPE OF LANDSLIDE:

The direction of slide is N40°W-S40°E. It is a deep-seated landslide. Sliding surface is mostly deeply located below the maximum rooting depth of trees especially at crown level up to middle level of slide. Field evidence suggests the involvement of deep regolith, weathered rock, and/or bedrock resulting into large slope failure associated with translational, rotational, or complex movement. Since it is debris-cum- rock landslide involving rapid movement, it is extremely destructive to Strategic Highway linking to Kashmir. The slide has following characteristics:

- The slope material in the landslide is mostly a debris heap, consisting of assorted, angular fragmental material with rich soil matrix.
- The bed rock is exposed at two locations, one at crown and another at right side at Road level. The bedrock along the roads is fragmented into angular boulders.
- The drainage along the Slide is bifurcated into two branches in the middle of slide.
- The blocks are dislodged from the crown area and brought down along with mud at road causing the rock –cum- debris fall.
- Rock- cum- debris fall is associated with rapid down slope movement of angular fragmental colluvial material and upper soil layer frequently causing huge blockade of the National Highway at Kheri.
- Debris flows are also common along the first or second order stream valleys in Muree group of rocks.

Table-3, SUMMARY OF FIELD MEASUREMENTS FROM THE KHERI LANDSLIDE SHOWING SLOPE ANGLE AND DEPTH OF SLIP SURFACE:

Survey Station (From)	Survey Station (To)	Length (m)	Width (m)	Slope angle (°)	Elevation (m)	Depth to slip surface (m)
20.0 m Below Crown	A (Near Highway)	A to F 600	120	20-27	820	8-10*
	B		130	30-35	819	5-7*
	C		115	26-35	845	3-5*
	D		85	35-40	885	5-7*
	E		90	28-32	867	4-6*
	F (Near crown)		80	44-70	1028	4-6*

(*Slip surface depth tentative at these locations/ unexposed due to slided material.)

Table-4, SUMMARY OF FIELD DATA MEASURED AT KHERI LANDSLIDE SHOWING DIP AND STRIKE:

Type	Strike	Amount	Dip
Bedding	N60°W–S60°E to N70°W – S 70°E	50° to 60°	Northerly.
Joint	N 35°W-S35°E	85°	Southerly.
Joint	N70°W-S70°E	65°	Due northerly.
Joint	N55°E-S55°W	80°	Northerly.

Table-5, SUMMARY OF FIELD ANALYSIS BASED ON WEATHERING AT KHERI LANDSLIDE:

No.	ZONES	DEGREE OF WEATHERING	DESCRIPTION
1	Toe	Slight	Discoloration on major discontinuity surfaces; rock material is discoloured and somewhat weaker than fresh rock.
2	Middle	Moderate	Less than half of the rock is present as a continuous framework and slightly discoloured.
3	Crown.	Severe	Most of rock material is decomposed; disintegrated to a soil, original mass structure is largely intact at Crown periphery.

Table showing the weathering degree of rocks (modified as per site condition at Kheri from Williamson 1984; Geological Society Engineering Working Party 1997; and Hoek and Bray 1997).

ELEVATION PROFILE OF KHERI:

The Y-axis of the chart displays the elevation, and the X-axis of the chart displays the distance (Fig.02). The % number displayed represents the % grade or slope which was calculated between top of Crown upto Toe for a distance of 674.0 m and the elevation difference is 163.0 m, where as maximum slope is 58° and average slope along profile is 46°.

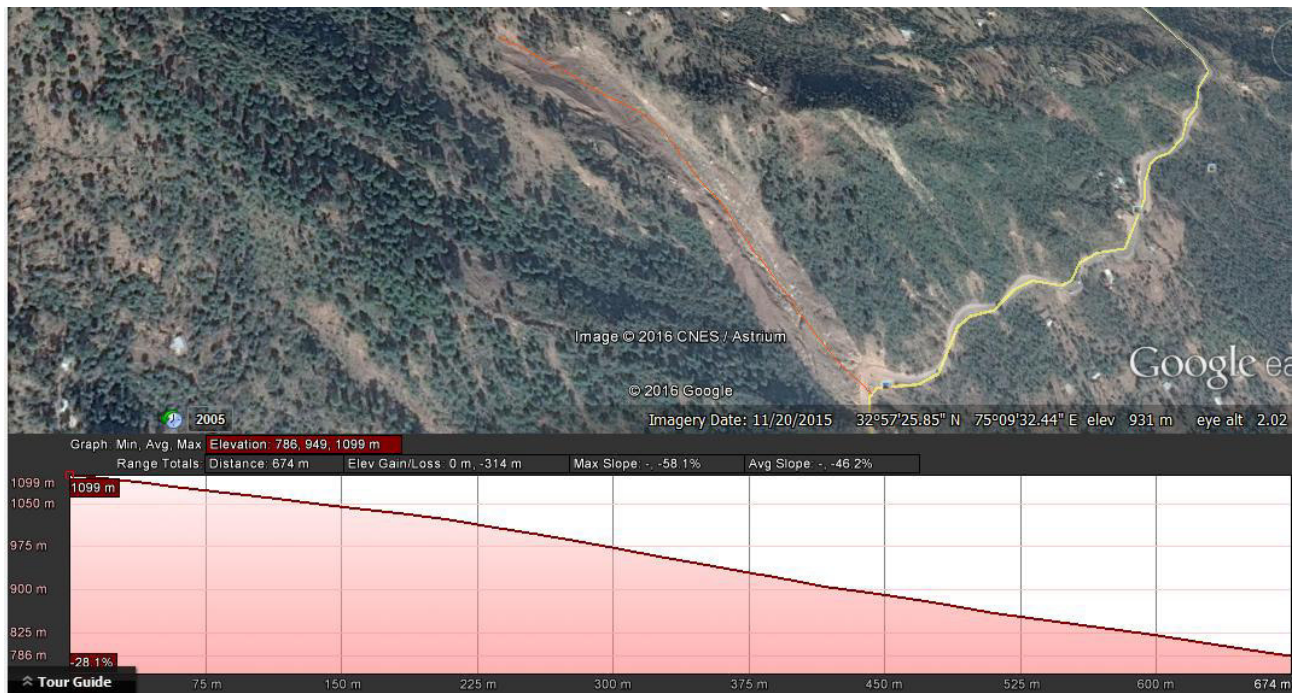


Fig. 02: Showing elevation profile along Kheri Slide.

LOCATION MAP OF STUDY AREA:

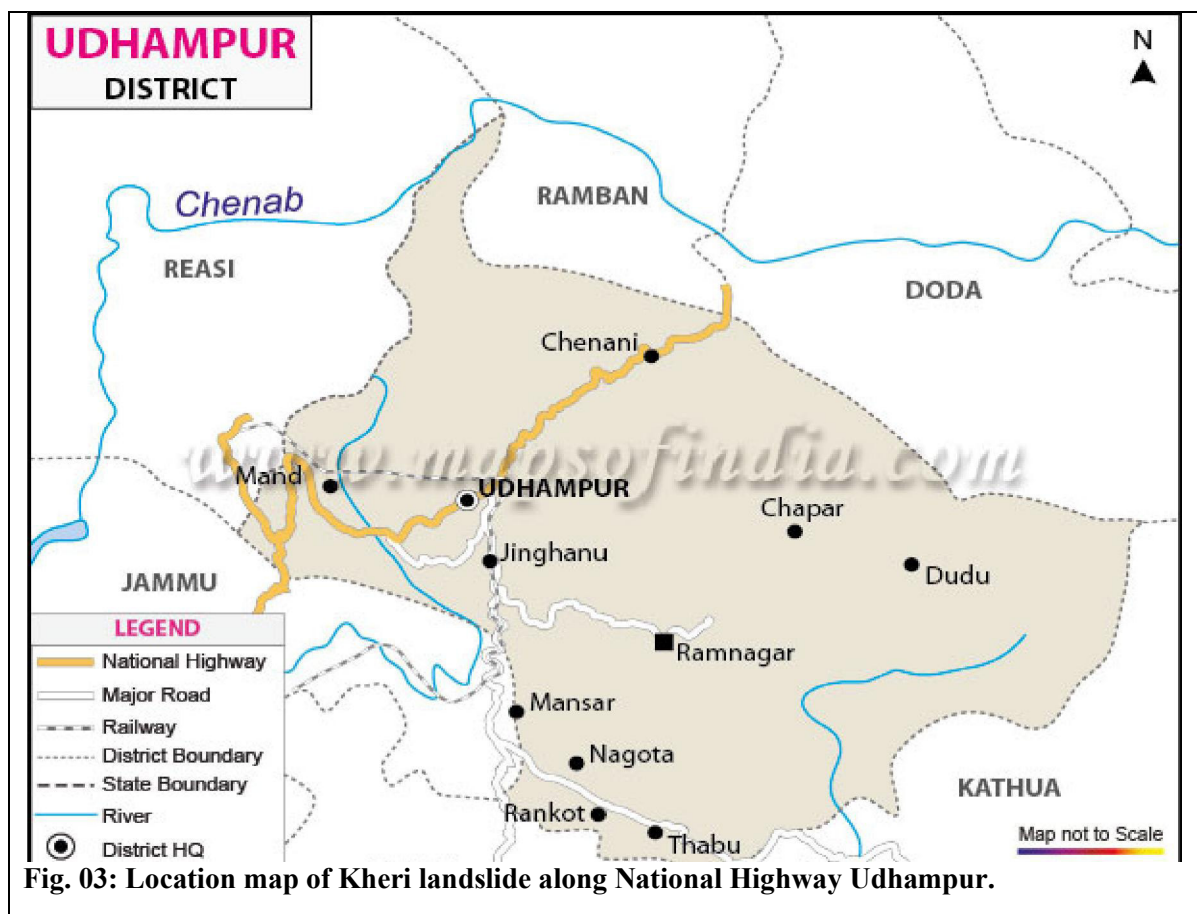
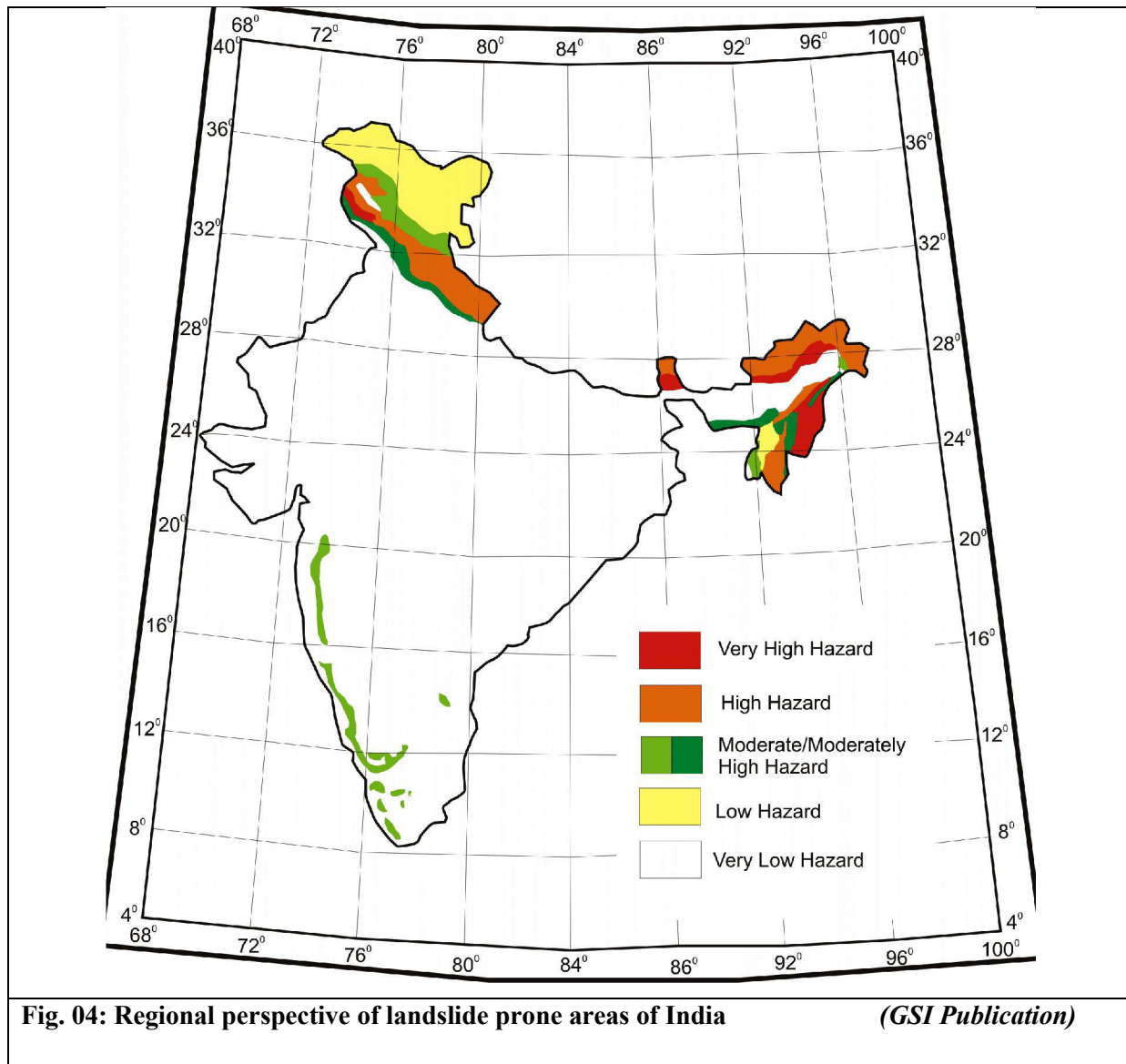


Fig. 03: Location map of Kheri landslide along National Highway Udhampur.



CAUSES OF KHERI LANDSLIDE:

Geologically, the area is occupied by the Muree Group of rocks, (Fig, 06) predominantly an argillaceous assemblage comprising alternate sequence of shale, siltstone, claystone and sandstone. Shale, siltstone, and claystone represent fragile and poorly consolidated rockmass, interbedded with hard, compact and fractured sandstone. The phenomenon of planar failure is very common in Muree rocks as shown in (Fig, 04) a very high hazard zone. Although on Compiled Geological map of the area, a major thrust is passing through north of slide, but continuous toe cutting at Road level, recharging of the debris by rain water, presence of one major nalla/Khad at crown level flowing through mid of slide (Fig, 08), coupled with absence of vegetation cover and high slope angle ranging from 40°-50°, are the main causative factors for the occurrence of Kheri landslide. Further, ponding of water (Fig, 06) in irregular depressions or

undrained areas on the hillslope above the Road section might have contributed to the gradual saturation of the debris resulting into the movement/sliding.

Bare, exposed, un-vegetated soil and debris is susceptible to movement/sliding even at the slightest rainfall, blocking the strategic route at regular intervals. The surrounding area of Kheri slide is potentially unstable, if left untreated.

The immediate triggering factors ascertained in the field that led to land sliding at particular spot, are heavy/intense rainfall during the summers (Fig.05) of 2015, erosion, and human factors especially at the toe. (Fig.07).

Unlined and improper drainage over the slide permits water to flow over an unprotected slope may also be a major factor in causing fresh landslides or recurrence of Landslide at Kheri. Water from these structures may saturate soils or severely erode the slope. Further, the presence of clay horizon and associated minerals significantly reduce the rock strength and facilitate the extensive failure of Kheri Slide. Clay-rich debris is also responsible for the formation of debris flows in the monsoon season.

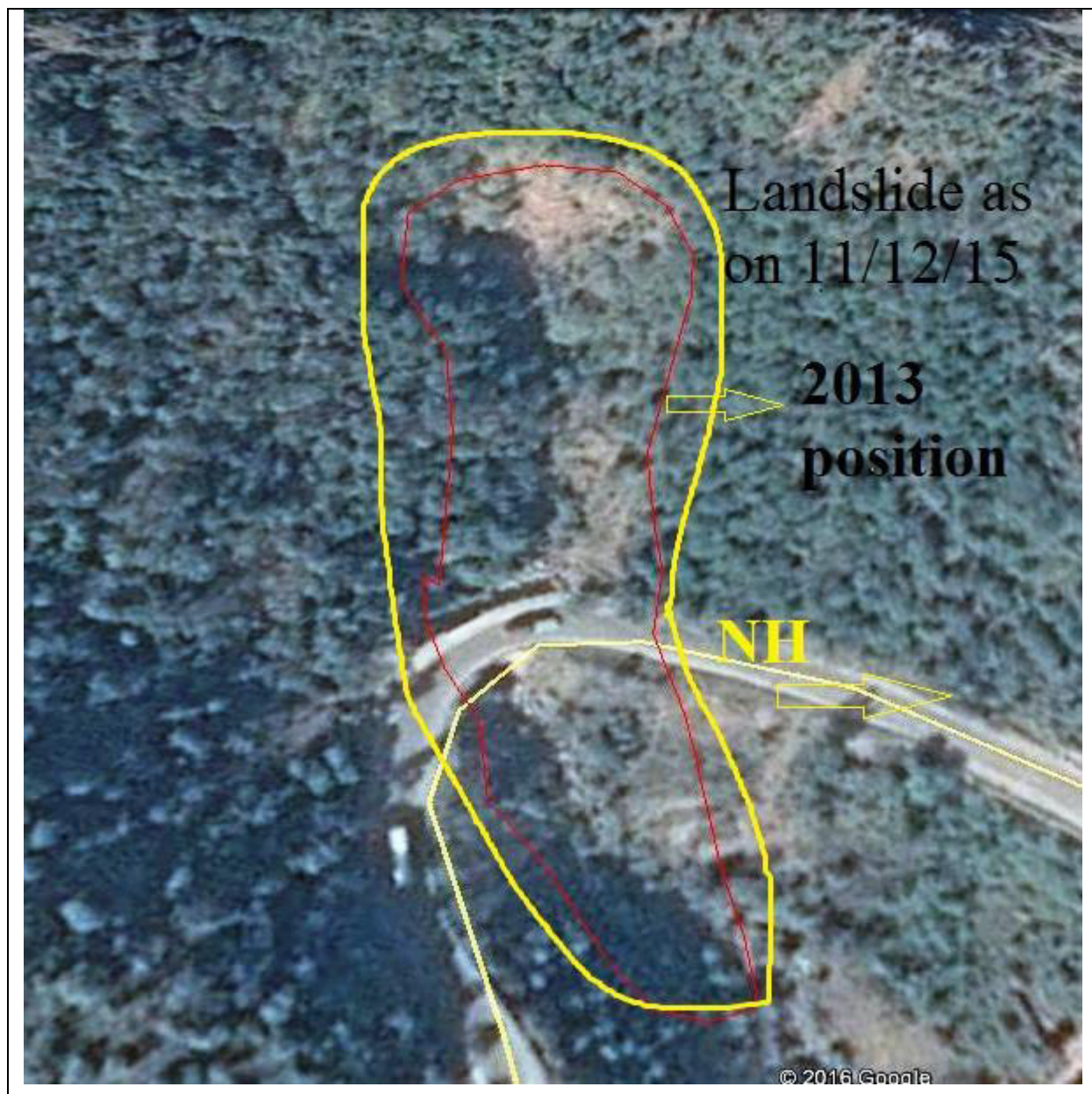


Fig. 05: Google image of Kheri landslide showing expansion.

IV- SUGGESTED REMEDIAL MEASURES:

Following remedial measures may help to stabilise the Kheri landslide:

- ❖ Drainage improvement in the area by means of contour and longitudinal drains.
- ❖ Toe cutting at road level should be avoided. (After widening permanent retaining wall should be given)
- ❖ Supporting wall at toe level and creation of benches along multiple locations of slide is needed.
- ❖ Nalla/Khad should be trained all along the slide.
- ❖ Soil nailing and vegetation may help to arrest the movement and eventually stabilize the slope in due course of time.
- ❖ Proper training of nala is required along with construction of gabion walls at suitable locations, Check bunds may be provided in the nalla at intermittent intervals.

The following alternatives for long term stability should also be considered:

- ❖ Reducing the height of the slope.
- ❖ Reducing the angle of the slope face at crown.
- ❖ Drainage of the slope.
- ❖ Planting vegetation along the slope.

BENCHING OF SLOPE OF KHERI SLIDE:

This method involves straight slopes separated by near horizontal bench. Benching increases stability of slopes by dividing the long slope into segments or smaller slopes connected by benches. The width of bench shall not be less than 8 m to enable the slope segments to act independently. In this method, construction becomes easier since steeper slopes are feasible with benches. The slope angle between the benches can be either uniform or variable. The benches shall be constructed with a V-shaped or gutter section with a longitudinal drainage grade and with suitable catch basins to carry the water down the slopes. The ditch shall be lined or paved to reduce erosion or to prevent percolation of water into pervious areas on the benches.

Field investigations indicate heavy inflow of surface and sub surface water from the uphill side above crown area to the landslide zone. Improvement in the drainage system with new drains and culverts has been suggested to divert the flow of water away from the landslide area.

RECOMMENDATION:

Geological and geotechnical investigations of landslide risk assessment involve a multidisciplinary approach. The investigations include preliminary stage geological investigations, detailed geological investigations and geotechnical investigations in this area where widening and four laning of Jammu Srinagar Highway is in progress with constant slope angle varying from 40° to 50° . The stability of this particular slope along highway from Udhampur to Ramban is critical because it is located mostly in Muree group of rocks, which is fragile in nature. Most of the landslides in the area are activated mainly during prolonged/excessive rainfall leading to widespread seepage and increase in pore water pressure. The potentially unstable slopes visible all along national highway need to be studied.

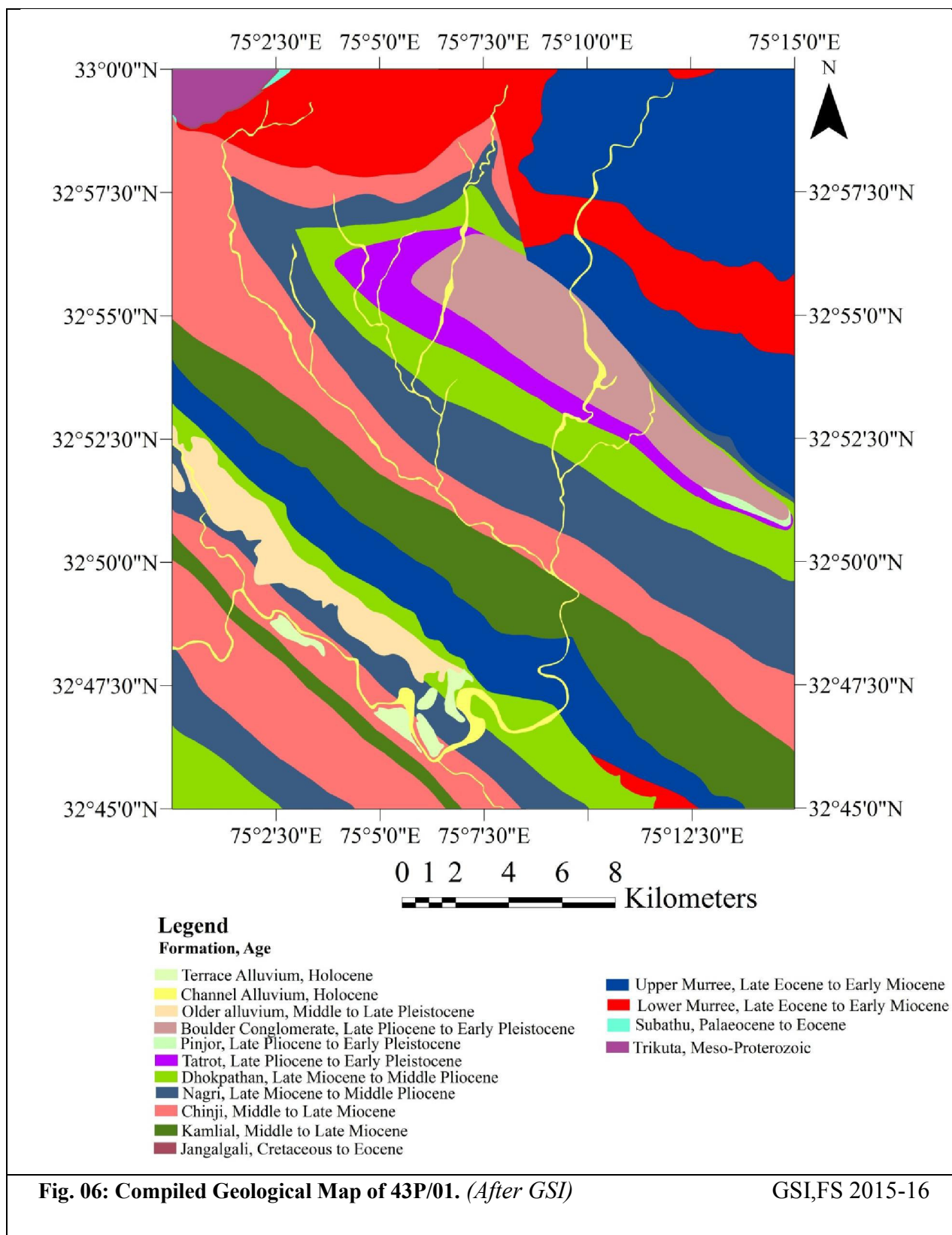




Fig. 07: Panoramic view of Kheri landslide.



Fig. 08: Showing location of Crown and Nalla/khad draining into crown of Kheri landslide.

LOCALITY INDEX

Locations	N (Degree)	E (Degree)
Kheri Slide	32°57'08"	75°09'18"
Jammu	32° 44' 00"	74° 52' 00"
Udhampur	32° 56' 0"	75°08' 00"
Katra	32°59' 30"	74°56' 00"
Patnitop pass	33°05'23"	75°19'43"
Chenani	33°20'51"	75°17'57"
Sud mahadev	33°01'14"	75°21'37"
Kud	33°44'47"	75° 17'52"
Man Talai	33°08'20"	75°21'30"
Sanna Sar	33°07'30"	75°11'40"

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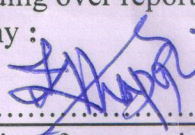
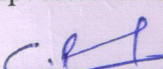
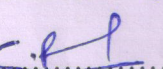
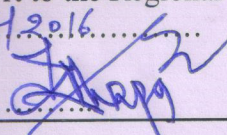
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Annexure-I

Geological Survey of India
Data sheet on report movement in connection with its scrutiny
Division: Engineering Geology and Landslide, FSP No: EG/C/NR/JK/2015/049,
Mission: IV

Title of the report: Report on detailed landslide investigation at Kheri, udhampur districts, Jammu & Kashmir

Name of author(s) Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist	Date of submission of report to the Supervisory Officer: 30/06/2016 Signature of the senior most author  30.06.16
Name of Supervisory Officer Sh. Janak Raj, Director	Date of receipt of report: 30/06/2016 Date of completion of scrutiny : Date of handing over report to the author after scrutiny : 21/07/2016 Signature 
Name of author(s) To whom the report has been handed over by the Supervisory Officer Devender Singh, Senior Geologist	Date of receipt after scrutiny 21/07/2016 Signature.....  Date of resubmission 27/07/2016 Signature..... 
Supervisory Officer Sh. Sh. Janak Raj, Director	Date of receipt after modification 27/07/2016 Date of handing over report to the Regional Mission Head.... 29/07/2016 Signature..... 
Name of Regional Mission Head Dr. Y. K. Sharma	Date of receipt of report 02/08/2016 Date of handing over the report after peer review to the Project Director 12/08/2016 Signature..... 

Name of Supervisory Officer Vijay Shivgotra Suptd. Geologist	Date of receipt from the Regional Mission Head after peer review 22/08/2016 Date of handing over to author(s) 23/08/2016 Signature..... Vijay Shivgotra
Name of author(s) Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist	Date of receipt of report from Supervisory Officer 23/08/2016 Date of handing over report after modification to Supervisory Officer 12/09/2016 Signature..... S.K.R.
Name of Supervisory Officer Vijay Shivgotra Suptd. Geologist	Date of receipt of report after modification from the author(s) 12/09/2016 Date of handing over report to Regional Mission Head 12/09/2016 Signature..... Vijay Shivgotra
Name of Regional Mission Head Deo Nath	Date of receipt after modification from the Project Director 16/09/2016 Date of handing over the report to the HoD of the Region for approval 20/09/2016 Signature..... Deo Nath 20/09/2016
Name of the HoD of the Region S.P. Nim	Date of receipt of report for approval 20-9-16 Date of approval..... 21-9-16 Date of sending back report to Project Director..... 21-9-16 Signature of the HoD. S.P. Nim
Name of the Project Director	Date of receipt of report after approval of HoD..... Date of circulation of report..... 30/09/2016 Signature.....



Annexure-II

Geological Survey of India

Certificate of Quality of the report

Project/Division: Engineering Geology and Landslide, FSP No: EG/C/NR/JK/2015/049, Mission: IV

Title of the report: Report on detailed landslide investigation at Kheri, udhampur districts, Jammu & Kashmir

Part-A

This is to certify that the following officer(s) have carried out the above mentioned FSP, and are responsible for collection of field data, carrying out the required laboratory analysis, compilation and synthesis of field data, coalition of the data and writing of the report as per the approved guidelines. The duration of field stay of each officer(s) is as against his/her name.

Name(s)/Designation of the Field Officer	Total field stay during the FSP (no. of days)	Signature with date
1. Devender Singh, Senior Geologist	14 Days	 12-09-16
2. Suresh Kumar Rana, Senior Geologist	14 Days	 12-09-16

Part-B

This is to certify that drilling has been carried out and completed as per the table given below:

Drilling Unit No	Date of commencement of drilling	Date of completion of drilling	Name and Signature of the Director Drilling
			NA

Part-C

This is to certify that geophysical component of the item has been carried out as per the table given below:

Name(s)/Designation of Geophysist	Date of commencement of work	Date of completion of work	Signature with date of the Geophysist
NA			
Name and signature with date of the Director (Geophysics)			

Part-D

This is to certify that the chemical analysis of the geological samples has been carried out as per the table given below:

Name of Director in charge of the Chemical Lab	Dates of receipt of samples from the field officer/party	Date of providing complete results to the field officer	Signature with date of the Director of the Chemistry
	NA	NA	NA

Part-E

This is to certify that the below mentioned Supervisory Officer(s) have supervised the above item and have provided guidance to the Field Officer(s). The number of inspection visits and total duration of field visits is as per the below table. It is also certified that the work carried out under the project is of high standards, the norms of field work, lab work etc. have been followed. It is also certified that thorough scrutiny of the report for maintaining the scientific contents and its Quality has been overseen.

Name(s)/Designation of the Supervisory	No. of inspection visits made	Total no. of days spent on	Signature with date
--	-------------------------------	----------------------------	---------------------

Officer	to the field area	inspection visit	
Sh. SK Tripathi, (The then Director) Sh. Rajesh Joshi, Director TC	0	NIL	<i>Rajesh Joshi</i> 19/09/2016 (RAJESH JOSHI)

Part-F

This is to certify that the report has been thoroughly scrutinized at the Regional Mission Head level. The scientific contents and the Quality of the report are certified. The report has been peer reviewed and the needful corrections have been attended to.

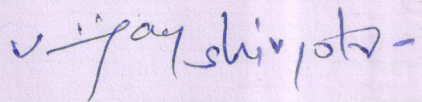
Name(s)/Designation of the Regional Mission Head	Signature with date
Deo Nath, Dy.D.G.	<i>Deo Nath</i> 20/09/2016

Part-G

This is to certify that the above report has been scrutinized at various levels and has also been peer reviewed. On being satisfied regarding the scientific contents and the Quality of the report, the undersigned has accorded approval for its circulation.

Name(s)/Designation of the Regional HoD	Signature with date
S. P. Nim, ADG & HOD	<i>S. P. Nim</i> 21/09/16

FIVE POINT PROFORMA

Title of the report	Report on Detailed Landslide Investigation at Kheri, Udhampur districts, Jammu & Kashmir.
Name(s) of the author(s)	Devender Singh , Sr. Geologist Suresh Kumar Rana, Sr. Geologist
Background of the investigation at whose instance or with what particular objective in view the investigation was undertaken	Since GSI is a nodal agency for landslides and as per guidelines by NMH-IV, by formal approval by HOD, NR, the Detailed Landslide investigation is initiated. During rainy season the Highway at Kheri get blocked due to landslide at many occasions throughout the year forcing authorities to halt the movement of vehicular traffic on this strategic route leading to Srinigar, Doda, Kashtwar etc.
Whether the item of investigation was included in the approved programme for the year concerned. If so, particulars, such as item No., page reference etc. may be given	Under Item No: (EG/C/NR/JK/2015/049) (Field Season: 2015-16),
Date of commencement of the investigation with month and year, and exact number of days spent by the officer(s) in the field in connection with the investigation.	FS: 2015-16 : 22/01/16 Devender Singh , Sr.Geologist (14 days) Suresh Kumar Rana, Sr.Geologist (14 days)
Brief comments by Director / Supervisory officer on the conclusion and recommendation by the author(s) with suggestion for implementing the same.	The conclusions and recommendations were made based on synthesis of field data.
Please state specifically how the results of the investigation were achieved compared with the objective for which it was undertaken	The results were achieved based on field data collection and interpretation.
Signature of the Project Director: 	Name: Vijay Shivgotra Designation: Superintending Geologist Project/ Division: EG & Landslide State Unit/ Region: Jammu & Kashmir, NR
Place	Jammu
Date	12-09-2016

REPORT DATA SHEET FOR PORTAL

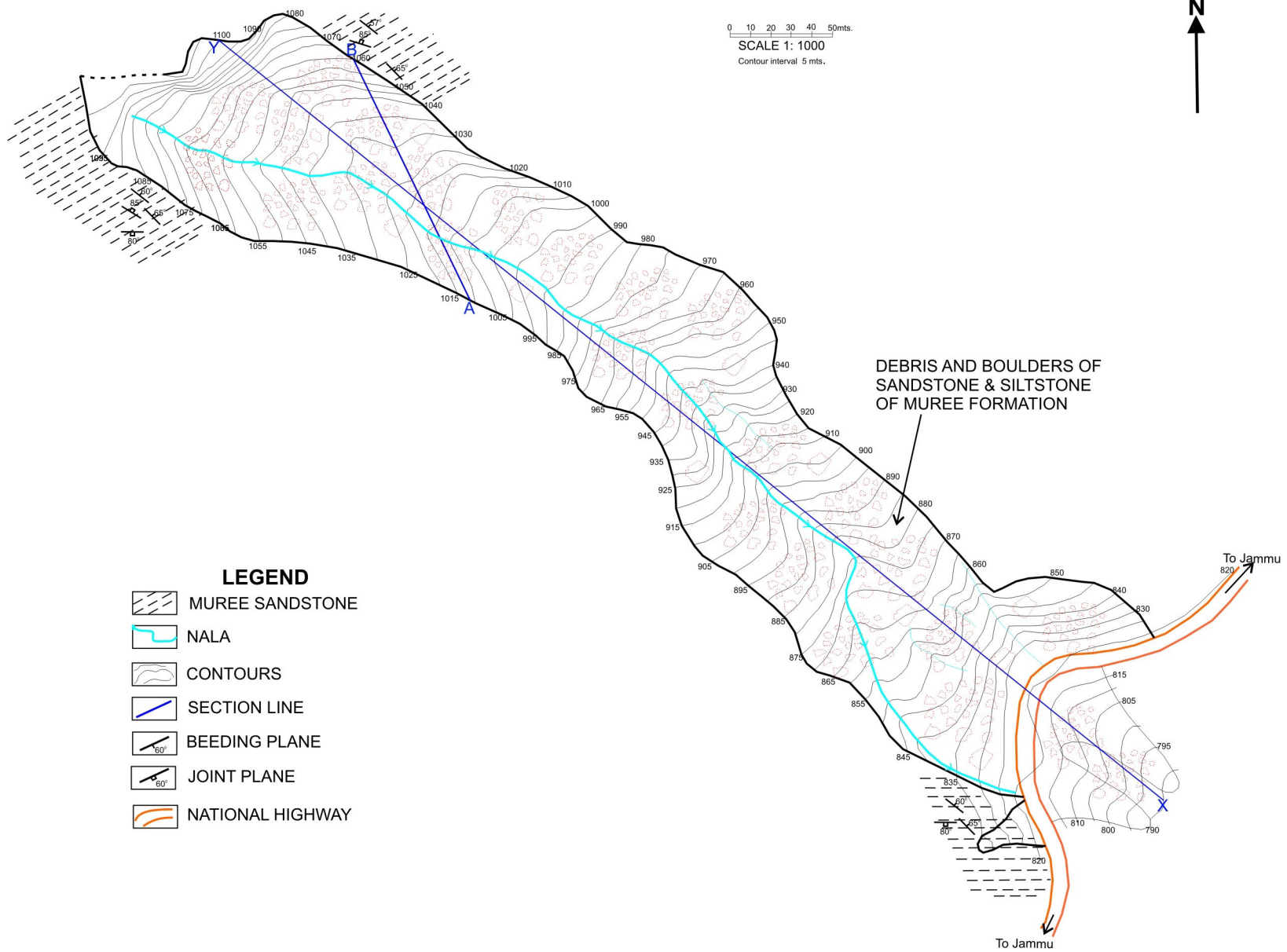
1	Title	Report on Detailed Landslide Investigation at Kheri, Udhampur districts, Jammu & Kashmir.
2	Author (s)	Devender Singh , Sr. Geologist Suresh Kumar Rana, Sr. Geologist
3	Field Season	FS: 2015-16
4	Year(s) (in case the item continues from an earlier FSP)	NA
5	FSP Code No.	Under Item No: (EG/C/NR/JK/2015/049)
6	Theme	Engineering Geology & Geotechnical Investigation
7	Subtheme	-
8	Status (Please refer to the letter attached)	-
9	State(s)	Jammu & Kashmir
10	District(s)	Udhampur
11	Toposheet	43P/01
12	Key Locations	Kheri, Udhampur
13	Short Abstract	Geological Survey of India has conducted detailed geotechnical assessment of the affected area and suggested remedial measures. Landslides along National highway occur in various forms and vary in mode, scale and frequency. The area falls in parts of SOI toposheet 43P/1 Udhampur, districts of J&K state. Udhampur is the main rail head which is part of a wider project to connect the Kashmir valley with the rest of India via a rail link. Moreover the area is connected by a network of roads, with main artery of road communication being the National Highway NH-44 which connects major towns like Udhampur, Chineni, Batote, Ramban, Ramsu and Banihal. NH-1B connects Batote Doda and Kishtwar along the Chenab River. Major drainage in the area is Tawi River; it is a principal tributary of the Chenab River and flows through its left banks. The flow of water in the river has been decreasing in recent years as the source glacier (Kali Kundi glacier) has been retreating. Detailed geotechnical assessment of the affected area was done and suggested remedial measures at landslide affected. Geologically, the area consists of folded and alternate sequence of purple to brick red shale, siltstone, and clay stone representing fragile and poorly consolidated rock mass, inter-bedded with buff purple hard, compact and fractured sandstone of the Muree Group of rocks. These rocks are well exposed along the slope as scanty patches and occur prominently at the ridges. The trend of the rock varies from N60°W – S60° E to N70° W – S 70° E with 50° to 60° northerly dips. The Kheri slide has occurred within Muree Group of rocks.

DETAILED GEOLOGICAL MAP OF KHERI LANDSLIDE, DISTRICT UDHAMPUR, J & K.

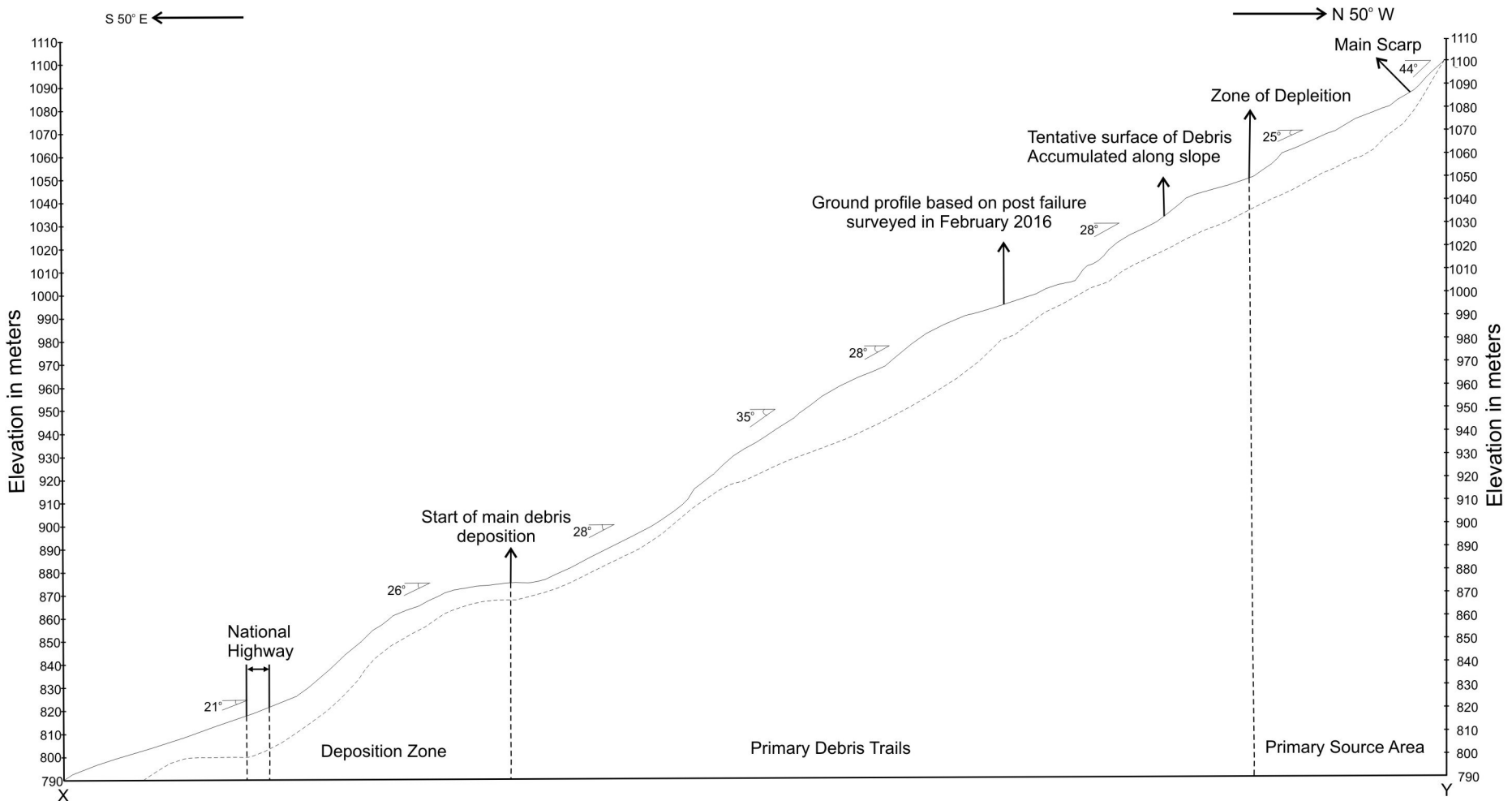


0 10 20 30 40 50mts.
SCALE 1: 1000
Contour interval 5 mts.

- LEGEND**
- MUREE SANDSTONE
 - NALA
 - CONTOURS
 - SECTION LINE
 - BEEDING PLANE
 - JOINT PLANE
 - NATIONAL HIGHWAY



GEOLOGICAL SECTION ALONG X - Y OF KHERI LANDSLIDE, DISTRICT UDHAMPUR, J & K



GEOLOGICAL SECTION ALONG A - B OF KHERI LANDSLIDE, DISTRICT UDHAMPUR, J & K

S 25° E ←

→ N 25° W

